

Cumulative Research Assignments

EDCI 6300

Part 1: Research Topic and Questions

Research Topic: Designing Artificial Intelligence to Support Rigorous Mathematics Learning for Students from Underprivileged Communities

Research Questions:

1. How can AI-driven learning systems be designed to assess prior knowledge and scaffold rigorous mathematics instruction for students from low-income communities?
2. To what extent do current AI educational tools reproduce existing opportunity gaps rather than address them?

Part 2: Literature Review

10 Peer-Reviewed Journal Articles (APA 7th Edition)

1. Walkington, C. (2025). The implications of generative artificial intelligence for mathematics education. *School Science and Mathematics*, 125(1), 3–15. <https://doi.org/10.1111/ssm.18356>
2. Ahmad, N., Murugesan, S., & Kshetri, N. (2023). Generative artificial intelligence and the education sector. *Computer*, 56(5), 113–122. <https://doi.org/10.1109/mc.2023.3263576>
3. Matzakos, N., Doukakis, S., & Moundridou, M. (2023). Learning mathematics with large language models. *International Journal of Emerging Technologies in Learning (iJET)*, 18(20), 4–21. <https://doi.org/10.3991/ijet.v18i20.42979>
4. Bui, P., Pongsakdi, N., McMullen, J., & Veermans, M. (2024). Affordances, challenges, and opportunities of ChatGPT in mathematics education: A scoping review. *Computers & Education*, 210, 104–118. <https://doi.org/10.31219/osf.io/ce2ky>
5. Thomas, D. R., Borchers, C., Kakarla, S., Lin, J., & Bhushan, S. (2025). Do tutors learn from equity training and can generative AI assess it? *Proceedings of the 15th International Conference on Learning Analytics & Knowledge*, 45–56. <https://doi.org/10.1145/3706468.3706531>
6. Price, M. D., Smith, E., & Smith, R. A. (2024). "Exceptional talent and enthusiasm for math": An examination of storylines circulated by ChatGPT about mathematical learners. *International Journal of Education in Mathematics Science and Technology*, 12(3), 789–804. <https://doi.org/10.46328/ijemst.4461>

7. Sapkota, B., & Bondurant, L. (2024). Assessing concepts, procedures, and cognitive demand of ChatGPT-generated mathematical tasks. *International Journal of Technology in Education*, 7(2), 245–262. <https://doi.org/10.46328/ijte.677>
8. Kaya, D., & Yavuz, S. (2025). Can generative AI and ChatGPT break human supremacy in mathematics and reshape competence in cognitive-demanding problem-solving tasks? *Journal of Intelligence*, 13(4), 43–58. <https://doi.org/10.3390/jintelligence13040043>
9. Smith, R. A., Smith, E., & Price, M. D. (2024). Utilizing emergent AI chatbot technology to generate mathematical writing models for elementary students with learning disabilities. *Intervention in School and Clinic*, 59(5), 312–320. <https://doi.org/10.1177/10534512241233512>
10. Biton, Y., & Segal, R. (2025). Learning to craft and critically evaluate prompts: The role of generative AI (ChatGPT) in enhancing pre-service mathematics teachers' TPACK and problem-posing skills. *International Journal of Education in Mathematics Science and Technology*, 13(1), 45–62. <https://doi.org/10.46328/ijemst.4654>

Current State of Knowledge

(1) Walkington (2025): This comprehensive review examines the implications of generative AI for mathematics education, focusing on four key areas: AI for mathematics problem-solving, AI for tutoring and feedback, AI to adapt mathematical tasks, and AI to assist mathematics teachers in planning. The study highlights both the potential benefits and ethical concerns of generative AI in educational contexts.

(2) Ahmad et al. (2023): This study explores how large language models like ChatGPT represent a significant breakthrough in conversational artificial intelligence and examines the varying institutional responses to LLMs in education. The authors discuss how some institutions ban LLMs while others encourage their use, highlighting the need for balanced approaches to AI integration.

(3) Matzakos et al. (2023): The researchers compare how large language models could be utilized in mathematics teaching compared to traditional computer algebra systems. They argue that LLMs provide functional and pedagogical opportunities that may influence changes in curriculum and assessment approaches in higher education mathematics.

(4) Bui et al. (2024): This scoping review of 28 studies examines the affordances, challenges, and opportunities of ChatGPT in mathematics education. The findings indicate that ChatGPT shows strengths in solving procedural and straightforward problems but has limitations with complex, multistep problems or those requiring visual and spatial reasoning.

(5) Thomas et al. (2025): This study evaluates whether tutors learn from equity training and whether generative AI can assess equity skills. The researchers found that both GPT-4o and GPT-4-turbo demonstrate proficiency in assessing tutors' ability to predict and explain the best approach when responding to students in potentially inequitable situations.

(6) Price et al. (2024): This exploratory study examines the storylines circulated by ChatGPT about mathematical learners from different racial/ethnic backgrounds and academic labels. The analysis revealed that ChatGPT participated in narratives where White and Asian students were portrayed as neutral while Black students were described as exceptional, potentially reinforcing problematic stereotypes.

(7) Sapkota & Bondurant (2024): The researchers investigated the procedures, concepts, and cognitive demand of ChatGPT-generated mathematical tasks focused on fraction multiplication. They found that despite ten variations of input prompts, ChatGPT did not produce tasks using the area model approach and instead generated mostly procedural tasks, some of which were conceptually inaccurate.

Synthesis of Common Trends and Contrasting Findings

The literature on generative AI in mathematics education reveals several consistent patterns. First, there is broad recognition that large language models represent a significant technological shift with potential to transform mathematics teaching and learning (Walkington, 2025; Ahmad et al., 2023). These models can engage in human-like conversation and generate answers to complex mathematical questions in real-time, offering new possibilities for personalized instruction and immediate feedback.

However, significant tensions emerge regarding the quality and appropriateness of AI-generated mathematical content. While Matzakos et al. (2023) highlight the functional and pedagogical opportunities of LLMs, Sapkota and Bondurant (2024) found that ChatGPT-generated mathematical tasks were mostly procedural, not cognitively demanding, and sometimes conceptually inaccurate. This contrast suggests that the sophistication of AI models does not guarantee pedagogical quality or alignment with reform-oriented mathematics education.

A third trend involves equity considerations in AI implementation. Thomas et al. (2025) demonstrate that generative AI can assess equity skills in tutors, suggesting potential for scaling equity training. However, Price et al. (2024) reveal concerning patterns in how ChatGPT circulates storylines about mathematical learners from different backgrounds, potentially reinforcing stereotypes. Bui et al. (2024) further note that ChatGPT's limitations with complex problems and tendency to produce verbose responses pose particular challenges for learners with limited mathematical proficiency, potentially widening rather than closing achievement gaps.

Theoretical Tendencies and Knowledge Gaps

The literature exhibits divergent theoretical tendencies between techno-optimism (emphasizing AI's transformative potential) and critical pedagogy (emphasizing how AI might

reproduce educational inequities). Walkington (2025) and Ahmad et al. (2023) represent the former, focusing on AI's capabilities and institutional responses. In contrast, Price et al. (2024) and Sapkota and Bondurant (2024) represent the latter, highlighting how AI-generated content can reinforce problematic narratives and produce pedagogically weak materials.

Several knowledge gaps emerge from this review. First, there is limited research on how generative AI can be designed specifically to support underprivileged students in mathematics. While Thomas et al. (2025) examine AI's ability to assess equity skills, and Smith et al. (2024) explore AI for students with learning disabilities, there is insufficient research on AI systems that intentionally scaffold mathematics learning for students from low-income communities with foundational gaps. Second, the literature lacks studies on how generative AI can assess and build on students' prior mathematical knowledge from their home and community contexts. Third, there is minimal research on hybrid human-AI models that combine the scalability of generative AI with the relational aspects of human teaching for marginalized students. Fourth, ethical considerations around algorithmic bias, data privacy, and the political economy of AI adoption in under-resourced schools remain under-explored. These gaps point to the need for research that examines not just whether generative AI works, but how it can be intentionally designed to disrupt rather than reproduce existing educational inequities for underprivileged mathematics learners.

Part 3: Research Methodology

The research methodology adopted for this study is qualitative research. Qualitative methodology is appropriate because both research questions are fundamentally concerned with understanding complex human and social phenomena rather than measuring or testing numerical outcomes. The first research question asks *how* AI-driven learning systems can be designed to assess prior knowledge and scaffold rigorous mathematics instruction for students from low-income communities. This is a design and process-oriented question that requires in-depth exploration of the experiences, perceptions, and needs of students, teachers, and designers — the kind of rich, contextual understanding that qualitative methods are uniquely positioned to provide. The second research question asks *to what extent* current AI tools reproduce opportunity gaps, which requires examining patterns of meaning, bias, and systemic inequity embedded in how these tools function and are experienced by marginalized learners — insights that cannot be fully captured through statistical measures alone.

Qualitative research is also appropriate because the topic itself — equity and AI in mathematics education for underprivileged communities — remains underexplored, and the field lacks a well-established theoretical or empirical framework to guide hypothesis testing. As the literature review reveals, significant knowledge gaps exist regarding how generative AI

systems are experienced by low-income students and how bias and inequity are reproduced through AI-generated content and design decisions. Qualitative methods such as interviews, observations, document analysis, and thematic analysis allow the researcher to explore these gaps inductively, surface the lived experiences of participants, and generate grounded theory that can later inform the design of more equitable AI tools. In this way, qualitative methodology aligns directly with both the exploratory nature of the research problem and the equity-centered commitments of the study.

Part 4: Research Design

The research design selected for this study is a **qualitative case study**. A case study design is the best fit for this research because it allows for an in-depth, context-sensitive examination of a specific, bounded phenomenon — in this instance, how generative AI tools function and are experienced within one or more real classrooms serving underprivileged students. Case study research is particularly well-suited to answering "how" and "to what extent" questions, both of which appear in this study's research questions. The first research question — how AI-driven systems can be designed to scaffold rigorous mathematics learning for students from low-income communities — requires the researcher to closely observe and analyze the actual interactions between students, teachers, and AI tools in context, rather than in a controlled or abstract setting. The second research question — to what extent current AI tools reproduce opportunity gaps — similarly demands a deep examination of real-world conditions, including the instructional decisions, institutional constraints, and student experiences that shape how AI is implemented in under-resourced schools. A case study design enables the collection of multiple forms of data, such as classroom observations, student and teacher interviews, and analysis of AI-generated content and prompts, which together provide a rich, triangulated understanding of the problem. Furthermore, because the literature review reveals that this topic is largely underexplored, a case study's inductive and exploratory nature makes it ideal for generating new insights and theoretical frameworks that can guide future research and the design of more equitable AI systems.